# Networking Fundamentals

Original introductory study example. Read each numbered lesson, then try the local demonstration quiz. No provider affiliation or professional certification is claimed.

## Chapter 1: Network foundations

### 1.1 What is a network?

A network connects devices so they can exchange information. A local area network covers a limited area.

### 1.2 Switches and routers

A switch connects devices within a LAN. A router forwards packets between networks.

## Chapter 2: Addresses and services

### 2.1 IPv4 addressing

An IPv4 address identifies an interface. The subnet mask defines the network portion.

### 2.2 DHCP and DNS

DHCP can assign network settings. DNS resolves names to records such as IP addresses.

## Chapter 3: Safe troubleshooting

### 3.1 Check from the connection outward

Check power, link status, addressing and the gateway before investigating a remote service.

### 3.2 Document the result

Record observations, approved changes and the outcome. Avoid changes you are not authorized to make.

## Practice checkpoint

Explain the concept in your own words, describe an approved example, and record what you still need to learn. The demonstration quiz includes single-answer, multiple-answer, fill-in-the-blank and matching questions.